

Feature Engineering Techniques for Overlap Reduction

1. Multiclass Classification Overlap Reduction

Technique 1.1: Linear Discriminant Analysis (LDA) Component Features

Mathematical Explanation & Equations

LDA projects high-dimensional features onto a lower-dimensional space while maximizing the ratio of between-class variance to within-class variance.

Let C be the number of classes. The between-class scatter matrix S_B and within-class scatter matrix S_W are defined as:

$$S_B = \sum_{c=1}^C N_c (\boldsymbol{\mu}_c - \boldsymbol{\mu})(\boldsymbol{\mu}_c - \boldsymbol{\mu})^T$$
$$S_W = \sum_{c=1}^C \sum_{\mathbf{x} \in C_c} (\mathbf{x} - \boldsymbol{\mu}_c)(\mathbf{x} - \boldsymbol{\mu}_c)^T$$

Where N_c is the number of samples in class c , $\boldsymbol{\mu}_c$ is the mean vector of class c , and $\boldsymbol{\mu}$ is the overall dataset mean vector.

LDA finds transformation vectors $\mathbf{W}$ that maximize Rayleigh's quotient:

$$J(\mathbf{W}) = \frac{\mathbf{W}^T S_B \mathbf{W}}{\mathbf{W}^T S_W \mathbf{W}}$$

The solution vectors are the eigenvectors corresponding to the top eigenvalues of $S_W^{-1} S_B$, yielding up to $C - 1$ new features.

Why to Use: It explicitly reshapes the feature space to pull cluster centroids apart while squeezing individual class clusters tight, directly reducing overlap.

When to Use: When class overlaps are driven by multi-feature linear interactions and features approximately follow a Gaussian distribution.

How to Use:

1. Fit an LDA model on training data using $C - 1$ components.
2. Transform both training and validation sets into $C - 1$ continuous features.
3. Append these projections as synthetic features alongside original variables.

Technique 1.2: Class Centroid Mahalanobis Distance Features

Mathematical Explanation & Equations

Calculates the distance from sample vector $\mathbf{x}$ to the centroid $\boldsymbol{\mu}_c$ of every class c , accounting for covariance structure $\boldsymbol{\Sigma}_c$:

$$D_M(\mathbf{x}, \boldsymbol{\mu}_c) = \sqrt{(\mathbf{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c)}$$

For C classes, this engineering step generates C continuous features: $[D_{M,1}, D_{M,2}, \dots, D_{M,C}]$.

Why to Use: Euclidean distance assumes spherical, isotropic class distributions. Mahalanobis distance scales along class covariance axes, separating overlapping classes that have distinct scale orientations or correlations.

When to Use: When overlapping classes exhibit distinct internal covariance structures or different spreads across multi-dimensional space.

How to Use:

1. Calculate the class mean vector $\boldsymbol{\mu}_c$ and covariance matrix $\boldsymbol{\Sigma}_c$ for each class c on training data.
2. Compute $D_M(\mathbf{x}, \boldsymbol{\mu}_c)$ for all samples across all classes $c \in \{1, \dots, C\}$.
3. Add these C generated continuous distance columns to the dataset.

Technique 1.3: Supervised Multiclass Out-of-Fold (OOF) Target Encoding

Mathematical Explanation & Equations

Converts high-cardinality categorical attributes into C continuous probability estimations using smoothed Bayesian posteriors.

For a categorical value k in sample i , the encoded feature for class c is:

$$S_{i,c} = \frac{\sum_{j \neq i} \mathbb{I}(y_j = c \wedge x_j = k) + m \cdot P(y = c)}{N_{k,-i} + m}$$

Where $\mathbb{I}(\cdot)$ is the indicator function, $N_{k,-i}$ is the count of category k excluding row i , $P(y = c)$ is the global prior probability of class c , and m is a smoothing weight parameter (e.g., $m = 10$).

Why to Use: Reduces overlapping categorical levels by mapping sparse categories directly to their multidimensional probability simplex vector.

When to Use: When categorical variables contain overlapping labels that correspond to different distributions across multiple target classes.

How to Use:

1. Split training data into K folds.
2. Calculate target class conditional probabilities strictly out-of-fold to prevent data leakage.
3. Create C distinct continuous columns for each categorical variable.

Technique 1.4: Soft-Clustering Gaussian Mixture Model (GMM) Features

Mathematical Explanation & Equations

Fits a GMM across the feature space to output soft component membership probabilities $P(z_k = 1 \mid \mathbf{x})$:

$$P(z_k = 1 \mid \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

Where $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ represents a multivariate normal distribution with mean $\boldsymbol{\mu}_k$ and covariance matrix $\boldsymbol{\Sigma}_k$.

Why to Use: Sub-clusters often exist within classes. GMM decomposition breaks multimodal, overlapping classes down into distinct sub-component densities.

When to Use: When classes are multimodal (e.g., Class 0 contains two separate geometric sub-clusters in space).

How to Use:

1. Train a GMM with K components ($K > C$) using unsupervised or semi-supervised methods.
2. Extract the K posterior probabilities for each sample.
3. Append these K soft cluster assignments as features.

2. Binary Classification Overlap Reduction

Technique 2.1: Weight of Evidence (WoE) Transformation

Mathematical Explanation & Equations

WoE transforms continuous or binned categorical attributes into a scaled measure of separation between positive and negative classes.

$$WoE_i = \ln \left(\frac{P(X = x_i | Y = 1)}{P(X = x_i | Y = 0)} \right) = \ln \left(\frac{\frac{\text{Positives}_i}{\text{Total Positives}}}{\frac{\text{Negatives}_i}{\text{Total Negatives}}} \right)$$

Associated Information Value (IV) validates feature effectiveness:

$$IV = \sum_{i=1}^B (P(X = x_i | Y = 1) - P(X = x_i | Y = 0)) \times WoE_i$$

Why to Use: Linearizes relationships between features and the log-odds of the target, forcing overlapping distribution tails to separate along a monotonic scale.

When to Use: When feature distributions between binary classes overlap heavily in non-linear or long-tailed patterns.

How to Use:

1. Discretize continuous variables into optimal quantile bins B .
2. Compute WoE_i for each bin i using training data.
3. Map feature values to their corresponding WoE scalar values.

Technique 2.2: Fisher's Linear Discriminant Projection

Mathematical Explanation & Equations

Reduces a multidimensional feature vector $\mathbf{x}$ to a 1D scalar projection y_{proj} maximizing binary class separation:

$$\mathbf{w} = \mathbf{S}_w^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$$

$$y_{proj} = \mathbf{w}^T \mathbf{x}$$

Where:

$$\mathbf{S}_w = \sum_{\mathbf{x} \in C_0} (\mathbf{x} - \boldsymbol{\mu}_0)(\mathbf{x} - \boldsymbol{\mu}_0)^T + \sum_{\mathbf{x} \in C_1} (\mathbf{x} - \boldsymbol{\mu}_1)(\mathbf{x} - \boldsymbol{\mu}_1)^T$$

Why to Use: It reduces multi-variable feature overlap by finding the single optimal 1D vector orthogonal to the decision boundary.

When to Use: When features are correlated, continuous, and show offset mean vectors between positive and negative classes.

How to Use:

1. Compute class mean vectors μ_0, μ_1 and within-class scatter matrix $\mathbf{S}_w$.
2. Compute vector weight $\mathbf{w}$.
3. Multiply input feature matrix $\mathbf{X}$ by $\mathbf{w}$ to create a 1D projection feature column.

Technique 2.3: k -Nearest Neighbor Class Density Ratio

Mathematical Explanation & Equations

Calculates the local positive-class probability density ratio around sample i using its k -nearest neighbors:

$$R_k(\mathbf{x}_i) = \frac{1}{k} \sum_{j \in N_k(\mathbf{x}_i)} \mathbb{I}(y_j = 1)$$

An optional distance-weighted variant is:

$$R_{k,w}(\mathbf{x}_i) = \frac{\sum_{j \in N_k(\mathbf{x}_i)} w_j \cdot \mathbb{I}(y_j = 1)}{\sum_{j \in N_k(\mathbf{x}_i)} w_j}, \quad w_j = \frac{1}{d(\mathbf{x}_i, \mathbf{x}_j) + \epsilon}$$

Why to Use: Directly addresses localized boundary overlap by providing explicit contextual metrics on surrounding neighborhood class composition.

When to Use: In non-linear, highly complex feature spaces where class boundaries are irregular or fragmented.

How to Use:

1. Fit a k -NN model using training features.
2. For each sample (using out-of-fold lookup for training sets), calculate the local ratio of positive neighbors.
3. Pass $R_k(\mathbf{x})$ as an input feature to the main downstream model.

Technique 2.4: Support Vector Margin Distance Features

Mathematical Explanation & Equations

Fits a Support Vector Classifier (SVC) with an RBF kernel to generate a signed margin distance feature $d_{margin}(\mathbf{x})$:

$$d_{margin}(\mathbf{x}) = \sum_{i \in SV} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$$

Where $K(\mathbf{x}_i, \mathbf{x}) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}\|^2)$ is the Radial Basis Function kernel and SV is the set of Support Vectors.

Why to Use: Kernelized SVM projects non-linearly overlapping features into high-dimensional space, transforming non-separable boundary overlap into clear linear distances from a hyper-plane.

When to Use: When linear transformations fail due to complex non-linear boundary shapes.

How to Use:

1. Train an SVC using a scaled RBF kernel on training data.
2. Extract decision boundary distances via `decision_function(X)`.
3. Add this signed distance as a continuous feature.

3. Regression Overlap & Non-Linear Mapping Techniques

Technique 3.1: Power Transformations (Box-Cox and Yeo-Johnson)

Mathematical Explanation & Equations

Stabilizes variance and normalizes skewed features or target distributions.

Box-Cox (requires $x > 0$):

$$x^{(\lambda)} = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln(x) & \text{if } \lambda = 0 \end{cases}$$

Yeo-Johnson (supports negative and zero values):

$$x^{(\lambda)} = \begin{cases} \frac{(x+1)^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0, x \geq 0 \\ \ln(x+1) & \text{if } \lambda = 0, x \geq 0 \\ -\frac{(-x+1)^{2-\lambda} - 1}{2-\lambda} & \text{if } \lambda \neq 2, x < 0 \\ -\ln(-x+1) & \text{if } \lambda = 2, x < 0 \end{cases}$$

Why to Use: In regression, “overlap” appears as unequal error variance (heteroscedasticity) across target ranges. Normalizing feature spaces minimizes density overlap across continuous target distributions.

When to Use: When features present heavily skewed tails or heteroscedastic residuals.

How to Use:

1. Estimate optimal parameter $\hat{\lambda}$ using Maximum Likelihood Estimation (MLE).
2. Apply transformation to the selected feature column.

Technique 3.2: Out-of-Fold (OOF) Target Mean & Variance Encoding for Regression

Mathematical Explanation & Equations

Encodes categorical features using target mean and target variance metrics derived out-of-fold.

Target Mean:

$$\mu_{i,S} = \frac{\sum_{j \neq i, x_j \in S} y_j + m \cdot \bar{y}_{global}}{|S_{-i}| + m}$$

Target Variance:

$$\sigma_{i,S}^2 = \frac{\sum_{j \neq i, x_j \in S} (y_j - \mu_{i,S})^2}{|S_{-i}| - 1}$$

Where S is the set of samples sharing the same categorical level, and m is the smoothing weight.

Why to Use: Replaces high-cardinality categorical variables with direct numeric indicators of target central tendency and uncertainty, collapsing wide overlapping distributions.

When to Use: When regression tasks include high-cardinality categorical attributes mapped to overlapping continuous response intervals.

How to Use:

1. Implement a K -fold cross-validation scheme.
2. Calculate target mean $\mu_{i,S}$ and target variance $\sigma_{i,S}^2$ strictly within training folds.
3. Map values to out-of-fold evaluation sets.

Technique 3.3: B-Splines and Radial Basis Function (RBF) Expansion

Mathematical Explanation & Equations

Decomposes a single feature x into a series of localized non-linear basis functions.

For M RBF kernels centered at knot locations c_m :

$$\phi_m(x) = \exp(-\gamma(x - c_m)^2), \quad m = 1, 2, \dots, M$$

B-Spline recursive basis definition:

$$B_{i,0}(x) = \begin{cases} 1 & \text{if } t_i \leq x < t_{i+1} \\ 0 & \text{otherwise} \end{cases}$$
$$B_{i,k}(x) = \frac{x - t_i}{t_{i+k} - t_i} B_{i,k-1}(x) + \frac{t_{i+k+1} - x}{t_{i+k+1} - t_{i+1}} B_{i+1,k-1}(x)$$

Where t_i represents knot locations.

Why to Use: Converts non-linear, multi-modal feature-target relationships into piecewise linear segments, resolving target variance overlaps across feature intervals.

When to Use: When a feature displays non-monotonic relationships with continuous targets (e.g., U-shaped or sinusoidal curves).

How to Use:

1. Select knot counts (M) using feature quantiles.
2. Generate M basis features per transformed column.
3. Replace original feature x with its set of expanded basis components $[\phi_1(x), \dots, \phi_M(x)]$.

Technique 3.4: k -NN Distance-Weighted Target Interpolation

Mathematical Explanation & Equations

Calculates expected continuous target values by averaging the targets of spatial nearest neighbors in the training set:

$$\hat{y}_{knn}(\mathbf{x}_i) = \frac{\sum_{j \in N_k(\mathbf{x}_i)} w_j \cdot y_j}{\sum_{j \in N_k(\mathbf{x}_i)} w_j}$$
$$w_j = \frac{1}{\|\mathbf{x}_i - \mathbf{x}_j\|_2 + \epsilon}$$

Why to Use: Directly maps complex local feature arrangements to expected target values, helping smooth noisy overlap zones across continuous response ranges.

When to Use: When high feature dimensionality causes localized target values to blur or overlap.

How to Use:

1. Normalize input features using Z-score standardization.
2. Fit a k -NN regressor on training feature space.
3. Generate out-of-fold target predictions $\hat{y}_{knn}$ and append them as a new feature.